Darren Huai

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EDUCATION

University of Southern California (USC) **Los Angeles, CA**
M.S. Computer Science July 2027
University of California, Los Angeles (UCLA) **Los Angeles, CA**
B.S. Mechanical/Aerospace Engineering: Concentration in Computer Science June 2024
Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence/Machine Learning, Neural Networks, Computer Networks, Software Architecture/Engineering, Data Science, Operating Systems, Programming System Design, Game Development

EXPERIENCE

National Aeronautics and Space Administration (NASA) **Edwards Air Force Base, CA**
Software Engineer Intern August 2026 - December 2026

- Built and tested real-time embedded systems software on NASA's **core Flight System (cFS)** for the **High-Performance Spaceflight Computing (HPSC)** processor, building **cFS apps and message-bus integrations in C** and automating **benchmarking and regression testing in Python** across **30+ configurations** to catch regressions pre-integration.
- Engineered **Python ETL pipelines** to ingest, validate, and process flight-test telemetry from NASA's **Research Aircraft for eVTOL Enabling Technologies (RAVEN)**, producing datasets and cutting manual processing time **45%** across **12 flights**.
- Developed **C++ software** within LabVIEW for fiber-optic systems supporting flight-test data collection and analysis

ITradeNetwork **Pleasanton, CA**
Software Engineer Intern June 2026 - August 2026

- Engineered a **full-stack AI agent** leveraging LLM tool-calling and API orchestration to automate software license lifecycle management across **500+ license accounts**, built with **React**, **Python**, and **SQL**, reducing manual effort by **40%**.
- Integrated **AI-assisted features** into an **enterprise SaaS platform** through **Python backend services** and **REST APIs**, enabling automated workflow processing across **5+ product modules** and cutting response times by **30%**.

Mercor (Series C AI Startup) **San Francisco, CA**
Software Engineer February 2025 - June 2026

- Developed **dynamic client-side web applications** using **React.js** and **RESTful APIs**, building reusable components and backend integrations that powered internal **RLHF tooling** and data workflows with **Gitlab CI/CD** pipelines.
- Engineered **RLHF pipelines and evaluation workflows** in **Python**, reducing processing time by **40%**
- Designed evaluation workflows measuring model accuracy on engineering and math tasks, improving benchmark accuracy by **18%** across **80+ model evaluations**.
- Evaluated and reviewed model responses on engineering problems spanning applied probability, game theory, and combinatorics, supporting benchmark accuracy toward **90%**

California Air Resources Board (CARB) **Riverside, CA**
Data Engineer Intern August 2023 - August 2024

- Refactored and optimized backend database infrastructure with **MySQL, XML, PHP, Drupal**, reducing query times by **2.5s.**

PROJECTS

ChessTan (Godot/GDScript) March 2026 - August 2026

- Built a custom client-server multiplayer backend in **Godot** for a 2-player hex-strategy game fusing settlement-based economy with chess-style combat, had a full rules engine (movement, capture, resource management) and a heuristic AI opponent
- Implemented host-authoritative real-time multiplayer over a custom WebSocket relay server with reconnect/resync support, and shipped cross-platform on **Discord Activities** and also **Steam**, with **8000+ downloads**

Autonomous Prediction Market Bot (Kalshi Bot) February 2026 - April 2026

- Built an autonomous bot in **Python** integrating **Kalshi's REST API with RSA private-key** authentication to scan **300+** live prediction markets (economic data, politics, etc.) every 5 minutes for pricing inefficiencies.
- Architected **DuckDB-backed** pipeline with **trade flow analysis** (taker-side imbalance, VWAP) and automated email alerts.

Generative Pre-Trained Transformer (Personal GPT) July 2025 - February 2026

- Developed a lightweight GPT model from scratch in **Python (PyTorch/NumPy)**, compressing core components (tokenization, self-attention, feed-forward layers) into **<200 LOC**, enabling rapid prototyping and educational deployment.
- Optimized training with gradient checkpointing, mixed precision, and cyclic learning rates, cutting GPU memory by **30%** and speeding training **20%** to reach **≈1.8** perplexity in **15 minutes** on an **RTX 4070**.

SKILLS

Programming Languages: Python, C++, SQL (PostgreSQL, MySQL), Javascript, MATLAB, Java, React, Svelte, C#, Node.js
Dev and Collaboration Tools: GitHub, Figma, Git, CI/CD, cFS, Linux, Docker, AWS, Pytest, Embedded Linux, Godot
Engineering Software: Excel, OpenRocket, Solidworks, Comsol, TensorFlow, Unreal Engine, LabVIEW, Matplotlib, DuckDB
Web Development/Data Analytics: HTML5, CSS3, GitHub Pages, Tableau, Bootstrap, PyTorch/NumPy, PHP, Rust, Ruby, Go